Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	06 Feb 2026
Team ID	LTVIP2026TMIDS84828
Project Name	Financial AI Analyzer – Transforming Complex Data into Actionable Insights
Maximum Marks	4 Marks

Functional Requirements

The following are the functional requirements of the proposed Financial AI Analyzer system.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Financial Data Upload	Upload financial datasets in CSV, Excel, or PDF format
		Validate file format before processing
		Display uploaded dataset preview
FR-2	AI-Based Financial Analysis	Convert uploaded data into structured format
		Send financial dataset to AI analysis module
		Generate structured financial insights and reports
		Display financial trends, charts, and key performance indicators
FR-3	Conversational Financial Assistant	Accept financial queries in natural language
		Convert query into structured prompt for AI processing
		Generate financial insights, recommendations, and explanations
		Restrict unrelated or unsupported queries

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-4	Structured Output Formatting	Present financial insights in structured format
		Display analysis using charts, tables, and categorized sections
		Provide summary insights and recommendations
FR-5	Error Handling	Show message for invalid file upload
		Display message if dataset cannot be processed
		Handle API or system errors gracefully
FR-6	Session State Management	Allow users to switch between Data Analysis Mode and Chat Assistant Mode
		Maintain analysis history during the session

Non-Functional Requirements

The following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application shall provide a simple and user-friendly interface with clear dashboards, financial charts, and easy navigation.
NFR-2	Security	The system shall securely manage API keys and protect financial data using secure storage and authentication mechanisms.
NFR-3	Reliability	The application shall handle multiple dataset uploads and financial queries without system crashes or interruptions.
NFR-4	Performance	The system shall generate financial insights within an average response time of **6–10 seconds** depending on dataset size.

NFR No.	Non-Functional Requirement	Description
NFR-5	Availability	The application shall be accessible via web deployment and available as long as the hosting platform and AI services remain active.
NFR-6	Scalability	The system shall support future enhancements such as real-time financial data integration, predictive analytics, multi-language support, and advanced AI-driven financial recommendations.